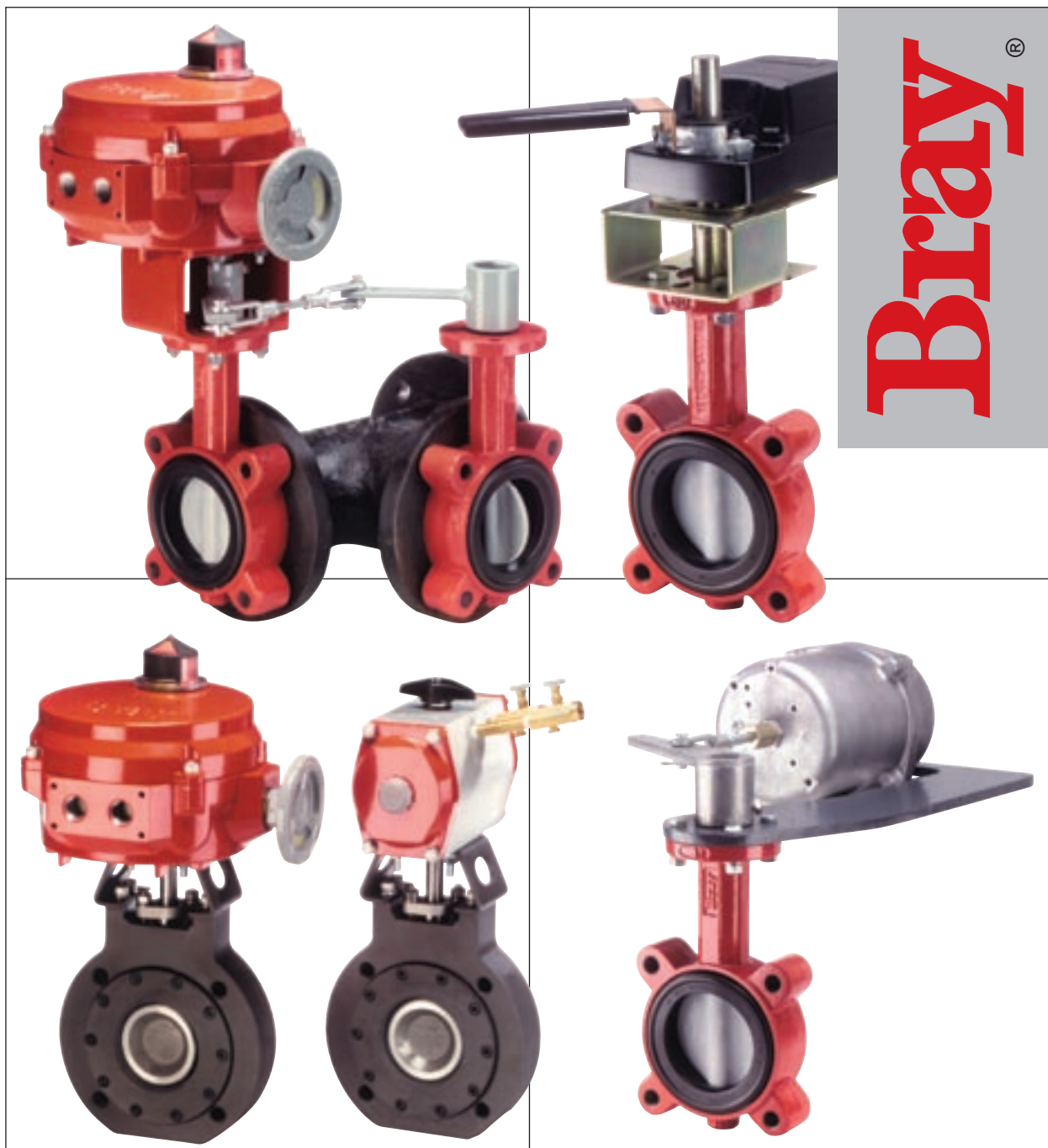


# Automated Butterfly Valves

**2-Way & 3-Way**

*Mixing & Diverting,  
Wafer and Lug Style Bodies  
For Hot & Chilled Water, Steam and Special Applications*



# Automated Butterfly Valves

## Features

### **SERIES 31**

**2"-20" Resilient Seated Valves**  
**Pressure Rating: 175 psi WOG**  
**Temp. Rating: -40°F to 250°F**  
**2 & 3-Way Assemblies**  
**Mixing and Diverting Services**

Bray Controls is proud to offer this automated line of butterfly valves. The Series 31 is available in a lug or wafer style body. Designed for general applications, this valve features long service life, ease of parts replacement, and interchangeability of components.

Bray's high quality automated butterfly valves precisely regulate the flow of hot and cold water or steam through all types of coils and heat exchangers.

The Series 31 2-Way valves have an equal percentage flow characteristic. The 3-Way valves have a liner flow. Both offer a very high rangeability and bubble tight shut-off capacity, and feature ease of installation and maintenance.

Optional Features Include:

- Aluminum Bronze discs
- Cast Steel bodies
- Fluoroelastomer Seats for high temperatures up to +400°F.

The Bray Series 31 offers a thin-profile disc without disc/stem fasteners to obstruct the flow. This design eliminates possible leak paths and allows for high  $C_v$  values with reduced turbulence. Stem design provides blow-out proof protection.

Most of Bray's electric and pneumatic actuators direct mount to Bray's butterfly valves, reducing space requirements.



### **ACTUATORS**

The low profile Series CA line of Electric Actuators operates on 120 or 220 VAC, and is available for use with incremental or proportional controllers. Proportional models feature a 0-10 VDC position feedback. Both spring return and non-spring return models are available.

The Series CA features an easily accessible manual override lever for manually positioning the valve when the actuator is not powered.

Bray offers additional actuators, both electric and pneumatic, to meet a variety of applications. Please see the Actuator section of this manual for more information.

### **SERIES 41**

**High Performance**  
**2-1/2"-20" ANSI Class 150**  
**Resilient Seated Valves**

The Series 41 offers bidirectional, zero leakage, shutoff at full rated ANSI Class 150 operating pressures. Rated 150 psi saturated steam at 366°F for on-off applications and 50 psi at 300°F for modulating service. The simple, innovative design offers rugged reliability and easy maintenance.

All statements, technical information, and recommendations in this bulletin are for general use only. Consult Bray representatives or factory for the specific requirements and material selection for your intended application. The right to change or modify product design or product without prior notice is reserved.

**Bray** **CONTROLS**  
**Commercial Division**

A Division of BRAY INTERNATIONAL, Inc.

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**CF Series**  
**Automated Butterfly Valves**  
**Valve Selection Data**

**C<sub>v</sub> VALUES – VALVE SIZING COEFFICIENT, SERIES 31**

| Valve Size | Valve Model No. | Disc Position In Degrees |     |      |      |      |      |       |       |       |
|------------|-----------------|--------------------------|-----|------|------|------|------|-------|-------|-------|
|            |                 | 10°                      | 20° | 30°  | 40°  | 50°  | 60°  | 70°   | 80°   | 90°   |
| 2"         | __LX-X020       | 1                        | 7   | 16   | 27   | 43   | 61   | 84    | 114   | 144   |
| 2-1/2"     | __LX-X025       | 1.5                      | 11  | 24   | 43   | 67   | 107  | 163   | 223   | 282   |
| 3"         | __LX-X030       | 2                        | 15  | 35   | 61   | 96   | 154  | 267   | 364   | 461   |
| 4"         | __LX-X040       | 3                        | 27  | 62   | 109  | 171  | 274  | 496   | 701   | 841   |
| 5"         | __LX-X050       | 5                        | 43  | 98   | 170  | 268  | 428  | 775   | 1146  | 1376  |
| 6"         | __LX-X060       | 6                        | 56  | 129  | 225  | 354  | 567  | 1025  | 1542  | 1850  |
| 8"         | __LX-X080       | 12                       | 102 | 241  | 421  | 680  | 1081 | 1862  | 2842  | 3316  |
| 10"        | __LX-X100       | 19                       | 162 | 382  | 667  | 1076 | 1710 | 2948  | 4525  | 5430  |
| 12"        | __LX-X120       | 27                       | 235 | 555  | 1005 | 1594 | 2563 | 4393  | 6731  | 8077  |
| 14"        | __LX-X140       | 34                       | 299 | 756  | 1320 | 2149 | 3384 | 5939  | 9974  | 10538 |
| 16"        | __LX-X160       | 45                       | 397 | 1001 | 1749 | 2847 | 4483 | 7867  | 11761 | 13966 |
| 18"        | __LX-X180       | 58                       | 507 | 1281 | 2237 | 3643 | 5736 | 10065 | 14496 | 17214 |
| 20"        | __LX-X200       | 72                       | 632 | 1595 | 2786 | 4536 | 7144 | 12535 | 18812 | 22339 |

C<sub>v</sub> is defined as the volume of water in U.S.G.P.M. that will flow through a given restriction or valve opening with a pressure drop of one (1) p.s.i. at room temperature.

**FLOW IN STANDARD WEIGHT PIPE AS A FUNCTION OF FLUID VELOCITY**

| Valve Size | Valve Model | Flow in Gallons Per Minute (GPM) |        |        |        |         |         |         |         |
|------------|-------------|----------------------------------|--------|--------|--------|---------|---------|---------|---------|
|            |             | 2 ft/s                           | 4 ft/s | 6 ft/s | 8 ft/s | 10 ft/s | 12 ft/s | 14 ft/s | 16 ft/s |
| 2"         | __LX-X020   | 21                               | 42     | 63     | 84     | 105     | 125     | 146     | 167     |
| 2 1/2"     | __LX-X025   | 30                               | 60     | 90     | 119    | 149     | 179     | 209     | 239     |
| 3"         | __LX-X030   | 46                               | 92     | 138    | 184    | 230     | 276     | 323     | 369     |
| 4"         | __LX-X04X   | 79                               | 159    | 238    | 317    | 397     | 476     | 555     | 635     |
| 5"         | __LX-X05X   | 125                              | 249    | 374    | 499    | 624     | 748     | 873     | 998     |
| 6"         | __LX-X06X   | 180                              | 360    | 540    | 720    | 900     | 1080    | 1261    | 1441    |
| 8"         | __LX-X08X   | 312                              | 624    | 939    | 1247   | 1559    | 1871    | 2183    | 2495    |
| 10"        | __LX-X10X   | 492                              | 983    | 1475   | 1966   | 2458    | 2949    | 3441    | 3932    |
| 12"        | __LX-X12X   | 705                              | 1410   | 2115   | 2820   | 3525    | 4230    | 4935    | 5640    |
| 14"        | __LX-X14X   | 859                              | 1719   | 2578   | 3438   | 4297    | 5157    | 6016    | 6876    |
| 16"        | __LX-X16X   | 1139                             | 2277   | 3416   | 4554   | 5693    | 6831    | 7970    | 9108    |
| 18"        | __LX-X18X   | 1457                             | 2914   | 4370   | 5827   | 7284    | 8741    | 10197   | 11654   |
| 20"        | __LX-X20X   | 1814                             | 3628   | 5442   | 7257   | 9071    | 10885   | 12699   | 14513   |

0=Full Rated Disc / 1=50 PSI Rated Disc  
 N=Normally Open / C=Normally Closed / 3-Way Configuration Number  
 2=2-Way / 3=3-Way  
 NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

## CF Series Automated Butterfly Valves

### Valve Selection Data

#### SERIES 31 VALVE WITH LOW PRESSURE ELECTRIC ACTUATORS

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |             |     |             |            |
|------------|-------------|-----------------------|-----------------------|------------------|-------------|-----|-------------|------------|
|            |             |                       |                       | NR/SR-2216       | NR-2224     | BBB | BIB         | BCA        |
| 2"         | __LX-X020   | 61                    | 44                    | 175              | 175         | 175 | 175         | 175        |
| 2-1/2"     | __LX-X025   | 107                   | 282                   | 100 / 175*       | 175         | 100 | 175         | 100 / 175* |
| 3"         | __LX-X030   | 154                   | 461                   | 175*             | 100 / 175*  | —   | 175         | 175*       |
| 4"         | __LX-X040   | 274                   | 841                   | 50(u)            | 50(u)/ 175* | —   | 50(u)/ 175* | 50(u)      |
| 5"         | __LX-X050   | 428                   | 1376                  | —                | 50(u)*      | —   | 50(u)*      | —          |
| 6"         | __LX-X060   | 567                   | 1850                  | —                | 50(u)*      | —   | 50(u)*      | —          |

#### SERIES 31 VALVE WITH SERIES 70 ELECTRIC ACTUATORS

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |         |         |         |         |         |         |
|------------|-------------|-----------------------|-----------------------|------------------|---------|---------|---------|---------|---------|---------|
|            |             |                       |                       | 70-0051          | 70-0081 | 70-0121 | 70-0201 | 70-0301 | 70-0501 | 70-0651 |
| 2"         | __LX-X020   | 61                    | 44                    | 175              | —       | —       | —       | —       | —       | —       |
| 2-1/2"     | __LX-X025   | 107                   | 282                   | 175              | —       | —       | —       | —       | —       | —       |
| 3"         | __LX-X030   | 154                   | 461                   | 175              | —       | —       | —       | —       | —       | —       |
| 4"         | __LX-X040   | 274                   | 841                   | 175              | —       | —       | —       | —       | —       | —       |
| 5"         | __LX-X050   | 428                   | 1376                  | 50(u)            | 175     | —       | —       | —       | —       | —       |
| 6"         | __LX-X060   | 567                   | 1850                  | —                | 175     | —       | —       | —       | —       | —       |
| 8"         | __LX-X080   | 1081                  | 3316                  | —                | 50(u)   | —       | 175     | —       | —       | —       |
| 10"        | __LX-X100   | 1710                  | 5430                  | —                | —       | —       | 50(u)   | 175     | —       | —       |
| 12"        | __LX-X120   | 2563                  | 8077                  | —                | —       | —       | —       | 50(u)   | 175     | —       |
| 14"        | __LX-X140   | 3384                  | 10538                 | —                | —       | —       | —       | —       | 50(u)   | 150     |
| 16"        | __LX-X160   | 4483                  | 13966                 | —                | —       | —       | —       | —       | —       | 50(u)   |
| 18"        | __LX-X180   | 5736                  | 17214                 | —                | —       | —       | —       | —       | —       | 50(u)   |

#### SERIES 31 VALVE WITH LOW PRESSURE PNEUMATIC ACTUATORS

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |        |        |          |
|------------|-------------|-----------------------|-----------------------|------------------|--------|--------|----------|
|            |             |                       |                       | D-3153           | D-3244 | D-3246 | D-3246-D |
| 2"         | __LX-X020   | 61                    | 44                    | 175              | —      | —      | —        |
| 2-1/2"     | __LX-X025   | 107                   | 282                   | 175              | —      | —      | —        |
| 3"         | __LX-X030   | 154                   | 461                   | 175              | —      | —      | —        |
| 4"         | __LX-X040   | 274                   | 841                   | 50(u)            | 175    | —      | —        |
| 5"         | __LX-X050   | 428                   | 1376                  | —                | 50(u)  | 175    | —        |
| 6"         | __LX-X060   | 567                   | 1850                  | —                | —      | 175    | —        |
| 8"         | __LX-X080   | 1081                  | 3316                  | —                | —      | 50(u)  | 175*     |
| 10"        | __LX-X100   | 1710                  | 5430                  | —                | —      | —      | 50(u)*   |

→ N=Normally Open / C=Normally Closed / 3-Way Configuration Number  
 → 2=2-Way / 3=3-Way  
 → NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\* Dual Actuators, (u)=undercut disc rated for bubble tight shut-off

**CF Series**  
**Automated Butterfly Valves**  
**Valve Selection Data**

**SERIES 31 VALVE WITH SERIES 92 DOUBLE ACTING PNEUMATIC ACTUATORS 80 PSI**

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |        |        |        |        |        |        |        |
|------------|-------------|-----------------------|-----------------------|------------------|--------|--------|--------|--------|--------|--------|--------|
|            |             |                       |                       | 92-063           | 92-083 | 92-092 | 92-118 | 92-127 | 92-160 | 92-210 | 90-255 |
| 2"         | __LX-X020   | 61                    | 44                    | 175              | 175    | —      | —      | —      | —      | —      | —      |
| 2-1/2"     | __LX-X025   | 107                   | 282                   | 175              | 175    | —      | —      | —      | —      | —      | —      |
| 3"         | __LX-X030   | 154                   | 461                   | 175              | 175    | —      | —      | —      | —      | —      | —      |
| 4"         | __LX-X040   | 274                   | 841                   | 50(u)            | 175    | —      | —      | —      | —      | —      | —      |
| 5"         | __LX-X050   | 428                   | 1376                  | —                | 150    | 175    | —      | —      | —      | —      | —      |
| 6"         | __LX-X060   | 567                   | 1850                  | —                | 50(u)  | 150    | 175    | —      | —      | —      | —      |
| 8"         | __LX-X080   | 1081                  | 3316                  | —                | —      | 50(u)  | 175    | 175    | —      | —      | —      |
| 10"        | __LX-X100   | 1710                  | 5430                  | —                | —      | —      | 50     | 150    | 175    | —      | —      |
| 12"        | __LX-X120   | 2563                  | 8077                  | —                | —      | —      | 50(u)  | 50(u)  | 175    | —      | —      |
| 14"        | __LX-X140   | 3384                  | 10538                 | —                | —      | —      | —      | —      | 100    | 150    | —      |
| 16"        | __LX-X160   | 4483                  | 13966                 | —                | —      | —      | —      | —      | 50(u)  | 150    | —      |
| 18"        | __LX-X180   | 5736                  | 17214                 | —                | —      | —      | —      | —      | 50(u)  | 100    | 150    |
| 20"        | __LX-X200   | 7144                  | 22339                 | —                | —      | —      | —      | —      | —      | 50(u)  | 150    |

→ N=Normally Open / C=Normally Closed / 3-Way Configuration Number  
 → 2=2-Way / 3=3-Way  
 → NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\*(u)=undercut disc rated for bubble tight shut-off

**SERIES 31 VALVE WITH SERIES 93 SPRING RETURN PNEUMATIC ACTUATORS 80 PSI**

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |         |          |          |          |          |          |
|------------|-------------|-----------------------|-----------------------|------------------|---------|----------|----------|----------|----------|----------|
|            |             |                       |                       | 93-83-4          | 93-92-4 | 93-118-4 | 93-127-4 | 93-160-4 | 93-210-4 | 91-255-4 |
| 2"         | __LX-X020   | 61                    | 44                    | 175              | —       | —        | —        | —        | —        | —        |
| 2 1/2"     | __LX-X025   | 107                   | 282                   | 175              | —       | —        | —        | —        | —        | —        |
| 3"         | __LX-X030   | 154                   | 461                   | —                | 175     | —        | —        | —        | —        | —        |
| 4"         | __LX-X040   | 274                   | 841                   | —                | 50(u)   | 175      | —        | —        | —        | —        |
| 5"         | __LX-X050   | 428                   | 1376                  | —                | —       | 50(u)    | 175      | —        | —        | —        |
| 6"         | __LX-X060   | 567                   | 1850                  | —                | —       | 50(u)    | 175      | —        | —        | —        |
| 8"         | __LX-X080   | 1081                  | 3316                  | —                | —       | —        | 50(u)    | 175      | —        | —        |
| 10"        | __LX-X100   | 1710                  | 5430                  | —                | —       | —        | —        | 50(u)    | 175      | —        |
| 12"        | __LX-X120   | 2563                  | 8077                  | —                | —       | —        | —        | —        | 50(u)    | 175      |
| 14"        | __LX-X140   | 3384                  | 10538                 | —                | —       | —        | —        | —        | 50(u)    | 175      |
| 16"        | __LX-X160   | 4483                  | 13966                 | —                | —       | —        | —        | —        | —        | 175      |
| 18"        | __LX-X180   | 5736                  | 17214                 | —                | —       | —        | —        | —        | —        | 50(u)    |
| 20"        | __LX-X200   | 7144                  | 22339                 | —                | —       | —        | —        | —        | —        | 50(u)    |

→ N=Normally Open / C=Normally Closed / 3-Way Configuration Number  
 → 2=2-Way / 3=3-Way  
 → NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\*(u)=undercut disc rated for bubble tight shut-off

**Note:** Actuators indicated are for 2-way, Normally Closed applications. For 3-way or Normally Open applications, consult your nearest Bray representative.

## CF Series Automated Butterfly Valves

### Valve Selection Data

#### SERIES 41 VALVE WITH SERIES 70 ELECTRIC ACTUATORS

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | 70-0051 | 70-0081 | 70-0121 | 70-0201 | 70-0301 | 70-0501 | 70-0651 |
|------------|-------------|-----------------------|-----------------------|---------|---------|---------|---------|---------|---------|---------|
| 2-1/2"     | MKL2-X025   | 78                    | 160                   | 150     | —       | —       | —       | —       | —       | —       |
| 3"         | MKL2-X030   | 123                   | 185                   | 150     | —       | —       | —       | —       | —       | —       |
| 4"         | MKL2-X040   | 250                   | 375                   | 150     | —       | —       | —       | —       | —       | —       |
| 5"         | MKL2-X050   | 360                   | 790                   | —       | —       | 150     | —       | —       | —       | —       |
| 6"         | MKL2-X060   | 510                   | 1350                  | —       | —       | 150     | —       | —       | —       | —       |
| 8"         | MKL2-X080   | 1060                  | 2800                  | —       | —       | —       | —       | 150     | —       | —       |
| 10"        | MKL2-X100   | 1630                  | 4300                  | —       | —       | —       | —       | —       | 150     | —       |
| 12"        | MKL2-X120   | 2530                  | 6650                  | —       | —       | —       | —       | —       | —       | 150     |

→ N=Normally Open / C=Normally Closed

#### SERIES 41 VALVE WITH SERIES 92 DOUBLE ACTING PNEUMATIC ACTUATORS 80PSI

| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |        |        |        |        |        |        |        |
|------------|-------------|-----------------------|-----------------------|------------------|--------|--------|--------|--------|--------|--------|--------|
|            |             |                       |                       | 92-063           | 92-083 | 92-092 | 92-118 | 92-127 | 92-160 | 92-210 | 90-255 |
| 2-1/2"     | MKL2-X025   | 78                    | 160                   | 150              | —      | —      | —      | —      | —      | —      | —      |
| 3"         | MKL2-X030   | 123                   | 185                   | 150              | —      | —      | —      | —      | —      | —      | —      |
| 4"         | MKL2-X040   | 250                   | 375                   | —                | 150    | —      | —      | —      | —      | —      | —      |
| 5"         | MKL2-X050   | 360                   | 790                   | —                | —      | 150    | —      | —      | —      | —      | —      |
| 6"         | MKL2-X060   | 510                   | 1350                  | —                | —      | 150    | —      | —      | —      | —      | —      |
| 8"         | MKL2-X080   | 1060                  | 2800                  | —                | —      | —      | 150    | —      | —      | —      | —      |
| 10"        | MKL2-X100   | 1630                  | 4300                  | —                | —      | —      | —      | —      | 150    | —      | —      |
| 12"        | MKL2-X120   | 2530                  | 6650                  | —                | —      | —      | —      | —      | 150    | —      | —      |

→ N=Normally Open / C=Normally Closed

#### SERIES 41 VALVE WITH SERIES 93 SPRING RETURN PNEUMATIC ACTUATORS 80PSI

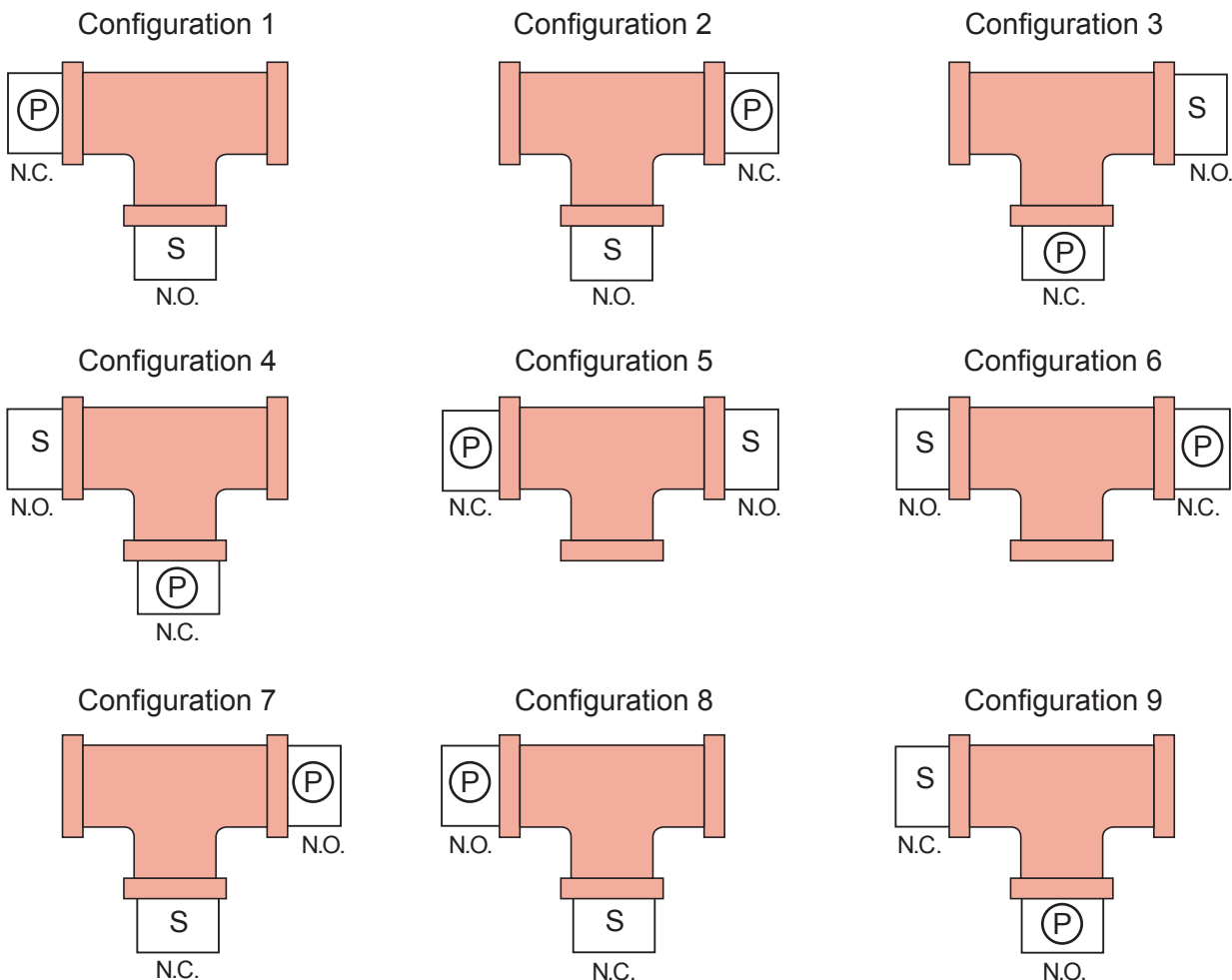
| Valve Size | Valve Model | C <sub>v</sub><br>60° | C <sub>v</sub><br>90° | Close-Off Rating |          |          |          |          |          |          |
|------------|-------------|-----------------------|-----------------------|------------------|----------|----------|----------|----------|----------|----------|
|            |             |                       |                       | 93-083-4         | 93-092-4 | 93-118-4 | 93-127-4 | 93-160-4 | 93-210-4 | 91-255-4 |
| 2-1/2"     | MKL2-X025   | 78                    | 160                   | 150              | —        | —        | —        | —        | —        | —        |
| 3"         | MKL2-X030   | 123                   | 185                   | —                | 150      | —        | —        | —        | —        | —        |
| 4"         | MKL2-X040   | 250                   | 375                   | —                | 150      | —        | —        | —        | —        | —        |
| 5"         | MKL2-X050   | 360                   | 790                   | —                | —        | —        | 150      | —        | —        | —        |
| 6"         | MKL2-X060   | 510                   | 1350                  | —                | —        | —        | 150      | —        | —        | —        |
| 8"         | MKL2-X080   | 1060                  | 2800                  | —                | —        | —        | —        | 150      | —        | —        |
| 10"        | MKL2-X100   | 1630                  | 4300                  | —                | —        | —        | —        | —        | 150      | —        |
| 12"        | MKL2-X120   | 2530                  | 6650                  | —                | —        | —        | —        | —        | —        | 150      |

→ N=Normally Open / C=Normally Closed

**Note:** Actuators indicated are for Normally Closed applications. For Normally Open applications, consult your nearest Bray representative.

**CF Series**  
**Automated Butterfly Valves**  
**3-Way Valve Flow**

**VALVE FLOW PATTERNS**



(P) Primary Valve/Actuator Location

S Secondary Valve

View is from top

Valves ordered without the valve configuration specified will be shipped as Configuration No. 3.

**Note:** All 3-Way butterfly valve assembly orders should have configuration specified. Pricing remains the same, however Bray must know the specifications in order to manufacture the appropriate linkage kit.

**Note:** 3-Way assemblies with low-pressure pneumatic actuators (D-series), are limited to arrangements: 1, 2, 7, and 8.

## CF Series Automated Butterfly Valves

### Dimensions

#### 2-WAY VALVE WITH CA SERIES ELECTRIC ACTUATOR

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D    | E    | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|------|------|------|------|------|------------------|-------|---------|
|              |              |                       |                       |      |      |      |      |      | BC               | Holes | Threads |
| 2"           | _ _ L2-X020  | 144                   | 61                    | 3.69 | 1.62 | 2.00 | 2.30 | 5.50 | 4.75             | 4     | 5/8-11  |
| 2-1/2"       | _ _ L2-X025  | 282                   | 107                   | 4.19 | 1.75 | 2.50 | 2.57 | 6.00 | 5.50             | 4     | 5/8-11  |
| 3"           | _ _ L2-X030  | 461                   | 154                   | 4.88 | 1.75 | 3.00 | 2.81 | 6.25 | 6.00             | 4     | 5/8-11  |
| 4"           | _ _ L2-X040  | 841                   | 274                   | 6.06 | 2.00 | 4.00 | 4.09 | 7.00 | 7.50             | 8     | 5/8-11  |
| 5"           | _ _ L2-X050  | 1376                  | 428                   | 7.06 | 2.12 | 5.00 | 4.61 | 7.50 | 8.50             | 8     | 3/4-10  |
| 6"           | _ _ L2-X060  | 1850                  | 567                   | 8.12 | 2.12 | 5.75 | 5.06 | 8.00 | 9.50             | 8     | 3/4-10  |

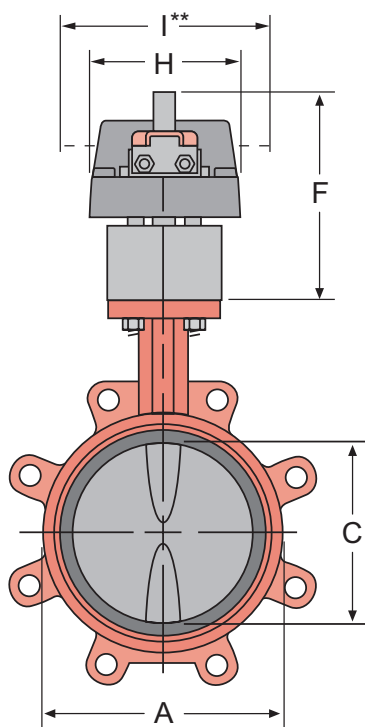
→ N=Normally Open / C=Normally Closed  
→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\*\*I = range of handle  
rotation

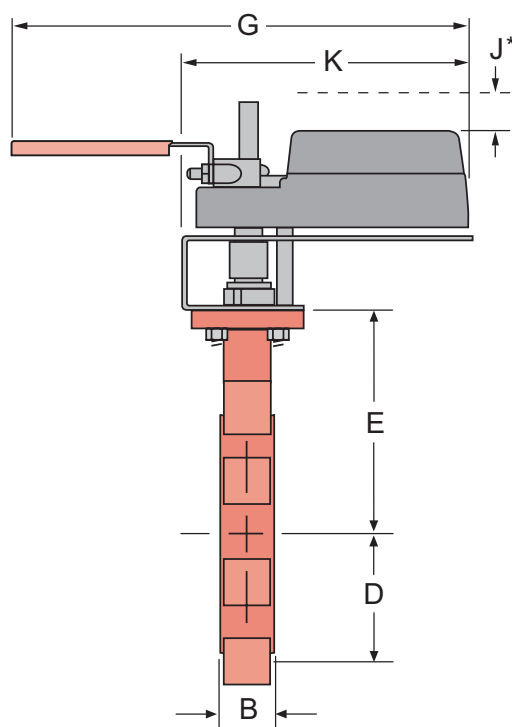
\*J = height required  
for cover removal

| Actuator<br>Model | F     | G     | H    | I**   | J*   | K     |
|-------------------|-------|-------|------|-------|------|-------|
| NR-2216           | 6.00  | 12.50 | 4.00 | 10.00 | 2.00 | 7.50  |
| NR-2224           | 6.00  | 12.50 | 4.00 | 10.00 | 2.00 | 7.50  |
| SR-2216           | 6.00  | —     | 4.00 | —     | 2.00 | 10.00 |
| B Series          | 6.00  | —     | 4.00 | —     | —    | 10.50 |
| Dual Act.         | 11.75 | 16.00 | 5.00 | —     | 2.00 | —     |

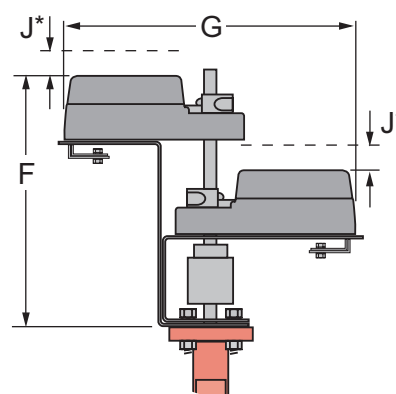
FRONT VIEW



SIDE VIEW



DUAL ACTUATORS



**CF Series**  
**Automated Butterfly Valves**  
**Dimensions**

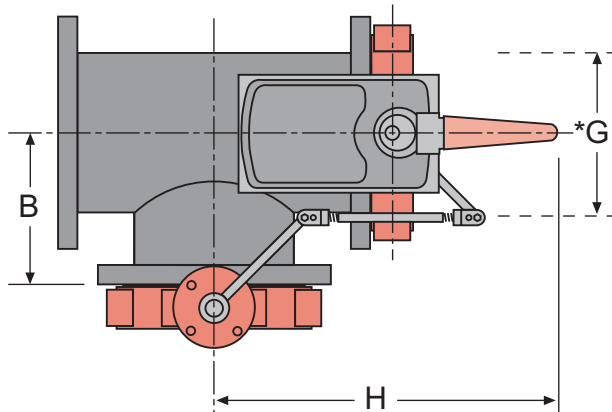
**3-WAY VALVE WITH CA SERIES ELECTRIC ACTUATOR**

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D    | E    | H     | Actuator<br>Model | F     | G *   |
|--------------|--------------|-----------------------|-----------------------|------|------|------|------|------|-------|-------------------|-------|-------|
| 2"           | _ _ L3-X020  | 144                   | 61                    | 2.00 | 4.50 | 3.00 | 1.62 | 5.50 | 12.00 | NR-2216           | 7.50  | 10.00 |
| 2-1/2"       | _ _ L3-X025  | 282                   | 107                   | 2.50 | 5.00 | 3.50 | 1.80 | 6.00 | 13.00 | NR-2224           | 7.50  | 10.00 |
| 3"           | _ _ L3-X030  | 461                   | 154                   | 3.00 | 5.50 | 3.80 | 1.80 | 6.20 | 13.50 | SR-2216           | 7.50  | —     |
| 4"           | _ _ L3-X040  | 841                   | 274                   | 4.00 | 6.50 | 4.50 | 2.00 | 7.00 | 14.50 | B Models          | 7.50  | —     |
| 5"           | _ _ L3-X050  | 1376                  | 428                   | 5.00 | 7.50 | 5.00 | 2.12 | 7.50 | 15.50 | Dual Act.         | 13.25 | —     |
| 6"           | _ _ L3-X060  | 1850                  | 567                   | 5.80 | 8.00 | 5.50 | 2.12 | 8.00 | 16.25 |                   |       |       |

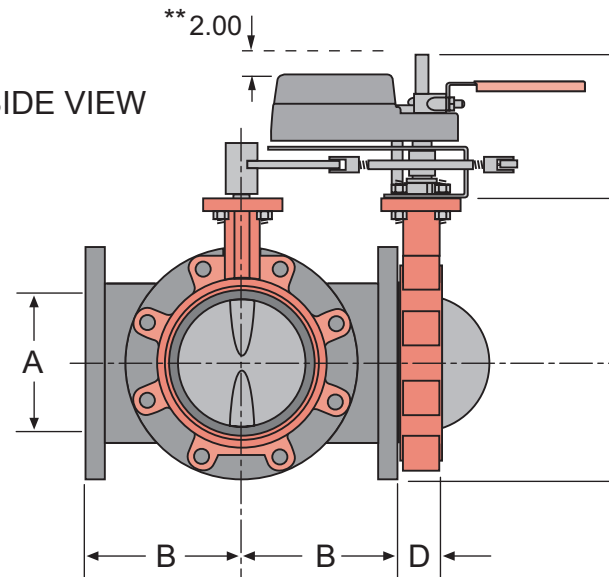
→ Configuration Number  
→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\*G = range of handle  
rotation

TOP VIEW

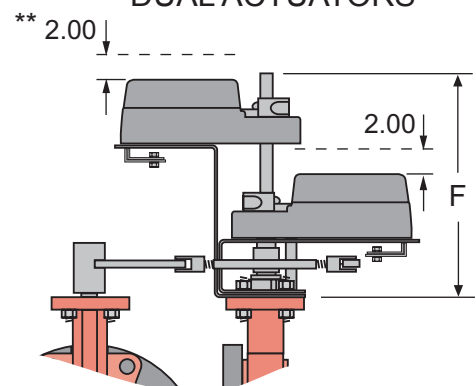


SIDE VIEW



\*\*minimum required  
for cover removal

DUAL ACTUATORS



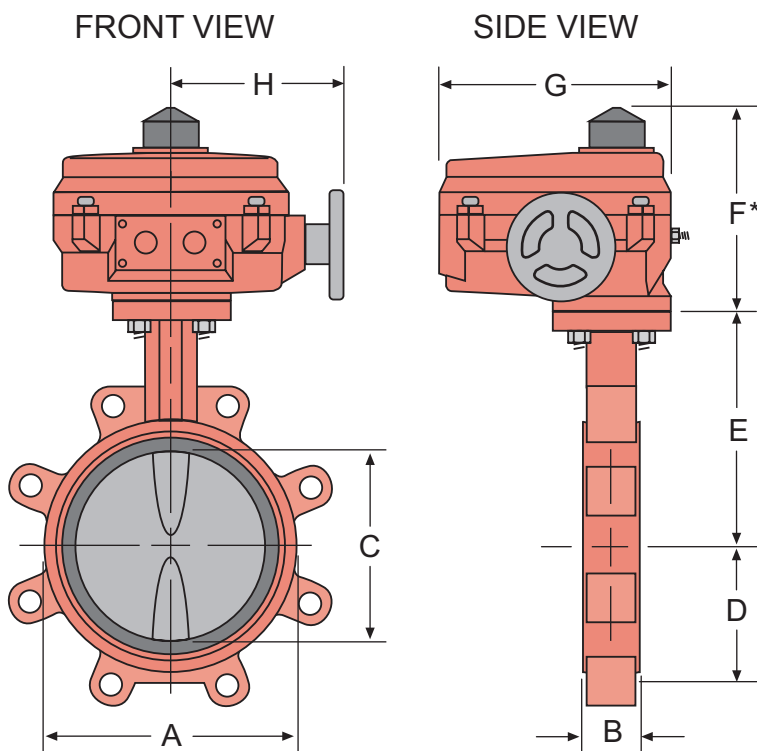
## CF Series Automated Butterfly Valves

### Dimensions

#### 2-WAY SERIES 31 VALVE WITH SERIES 70 ELECTRIC ACTUATOR

| Valve<br>NPT | Model<br>No. | $C_v$ |      | A     | B    | C     | D     | E     | Lug Bolting Data |       |         |
|--------------|--------------|-------|------|-------|------|-------|-------|-------|------------------|-------|---------|
|              |              | 90°   | 60°  |       |      |       |       |       | BC               | Holes | Threads |
| 2"           | __L2-X020    | 144   | 61   | 3.69  | 1.62 | 2.00  | 2.30  | 5.50  | 4.75             | 4     | 5/8-11  |
| 2-1/2"       | __L2-X025    | 282   | 107  | 4.19  | 1.75 | 2.50  | 2.57  | 6.00  | 5.50             | 4     | 5/8-11  |
| 3"           | __L2-X030    | 461   | 154  | 4.88  | 1.75 | 3.00  | 2.81  | 6.25  | 6.00             | 4     | 5/8-11  |
| 4"           | __L2-X040    | 841   | 274  | 6.06  | 2.00 | 4.00  | 4.09  | 7.00  | 7.50             | 8     | 5/8-11  |
| 5"           | __L2-X050    | 1376  | 428  | 7.06  | 2.12 | 5.00  | 4.61  | 7.50  | 8.50             | 8     | 3/4-10  |
| 6"           | __L2-X060    | 1850  | 567  | 8.12  | 2.12 | 5.75  | 5.06  | 8.00  | 9.50             | 8     | 3/4-10  |
| 8"           | __L2-X080    | 3316  | 1081 | 10.50 | 2.50 | 7.75  | 6.05  | 9.50  | 11.75            | 8     | 3/4-10  |
| 10"          | __L2-X100    | 5430  | 1710 | 12.75 | 2.50 | 9.75  | 7.69  | 10.75 | 14.25            | 12    | 7/8-9   |
| 12"          | __L2-X120    | 8077  | 2563 | 14.88 | 3.00 | 11.75 | 9.02  | 12.25 | 17.00            | 12    | 7/8-9   |
| 14"          | __L2-X140    | 10538 | 3384 | 17.05 | 3.00 | 13.25 | 9.93  | 13.62 | 18.75            | 12    | 1-8     |
| 16"          | __L2-X160    | 13966 | 4483 | 19.21 | 4.00 | 15.25 | 11.30 | 14.75 | 21.25            | 16    | 1-8     |
| 18"          | __L2-X180    | 17214 | 5736 | 21.12 | 4.25 | 17.25 | 12.16 | 16.00 | 22.75            | 16    | 1 1/8-7 |

→ N=Normally Open / C=Normally Closed  
 → NY=Nylon Coated Disc / AB=Aluminum Bronze Disc



| Actuator<br>Model | F *  | G     | H    |
|-------------------|------|-------|------|
| 70-0051           | 6.70 | 7.50  | 5.80 |
| 70-0081/0121/0201 | 8.10 | 10.10 | 7.80 |
| 70-0301/0501/0651 | 8.80 | 12.10 | 9.50 |

\* Allow additional 5" for Actuator Cover removal.

**CF Series**  
**Automated Butterfly Valves**  
**Dimensions**

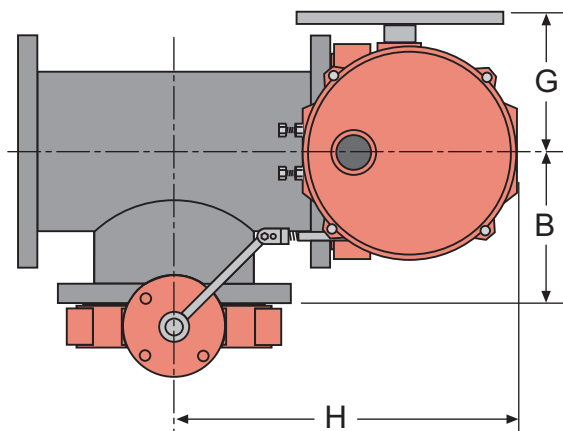
**3-WAY SERIES 31 VALVE WITH SERIES 70 ELECTRIC ACTUATOR**

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A     | B     | C     | D    | E     | Actuator<br>Model | F *   | G    | H     |
|--------------|--------------|-----------------------|-----------------------|-------|-------|-------|------|-------|-------------------|-------|------|-------|
| 2"           | _ _L3-X020   | 144                   | 61                    | 2.00  | 4.50  | 3.00  | 1.62 | 5.50  | 70-0051           | 9.70  | 5.80 | 11.00 |
| 2-1/2"       | _ _L3-X025   | 282                   | 107                   | 2.50  | 5.00  | 3.50  | 1.80 | 6.00  | 70-0051           | 9.70  | 5.80 | 11.60 |
| 3"           | _ _L3-X030   | 461                   | 154                   | 3.00  | 5.50  | 3.80  | 1.80 | 6.20  | 70-0051           | 9.70  | 5.80 | 12.00 |
| 4"           | _ _L3-X040   | 841                   | 274                   | 4.00  | 6.50  | 4.50  | 2.00 | 7.00  | 70-0081           | 11.10 | 7.80 | 15.20 |
| 5"           | _ _L3-X050   | 1376                  | 428                   | 5.00  | 7.50  | 5.00  | 2.12 | 7.50  | 70-0121           | 11.10 | 7.80 | 16.30 |
| 6"           | _ _L3-X060   | 1850                  | 567                   | 5.80  | 8.00  | 5.50  | 2.12 | 8.00  | 70-0201           | 11.10 | 7.80 | 16.80 |
| 8"           | _ _L3-X080   | 3316                  | 1081                  | 7.80  | 9.00  | 6.80  | 2.50 | 9.50  | 70-0301           | 12.80 | 9.50 | 19.50 |
| 10"          | _ _L3-X100   | 5430                  | 1710                  | 9.80  | 11.00 | 8.00  | 2.50 | 10.80 | 70-0501           | 12.80 | 9.50 | 21.50 |
| 12"          | _ _L3-X120   | 8077                  | 2563                  | 11.80 | 12.00 | 9.50  | 3.00 | 12.20 | 70-0651           | 12.80 | 9.50 | 22.70 |
| 14"          | _ _L3-X140   | 10538                 | 3384                  | 13.20 | 14.00 | 10.50 | 3.00 | 13.60 | 70-0651           | 12.80 | 9.50 | 24.70 |
| 16"          | _ _L3-X160   | 22339                 | 7144                  | 15.20 | 15.00 | 11.80 | 4.00 | 14.80 | 70-0651           | 12.80 | 9.50 | 26.20 |

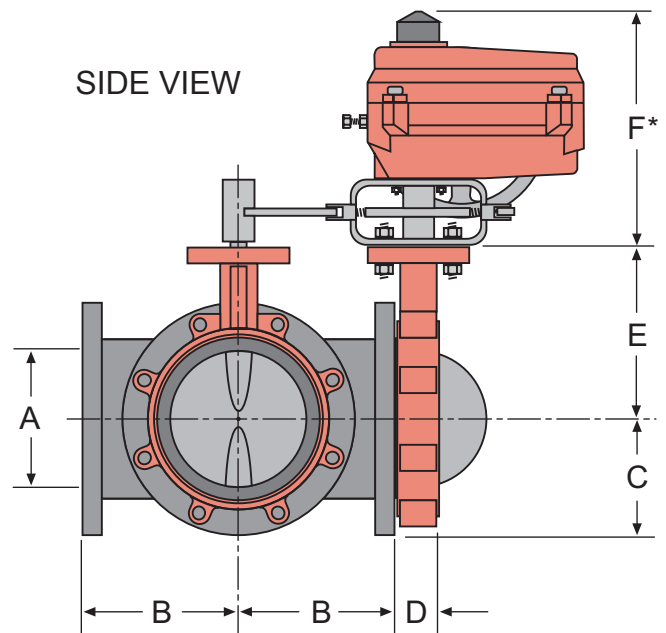
Configuration Number  
NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

\* Allow additional 5" for Actuator  
Cover removal.

TOP VIEW



SIDE VIEW



## CF Series Automated Butterfly Valves

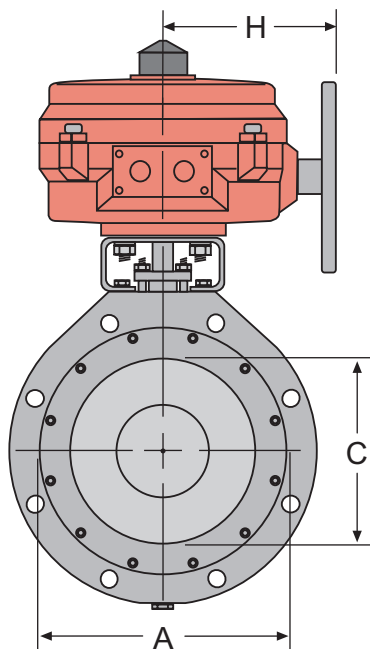
### Dimensions

#### 2-WAY SERIES 41 HIGH PERFORMANCE VALVE WITH SERIES 70 ELECTRIC ACTUATOR

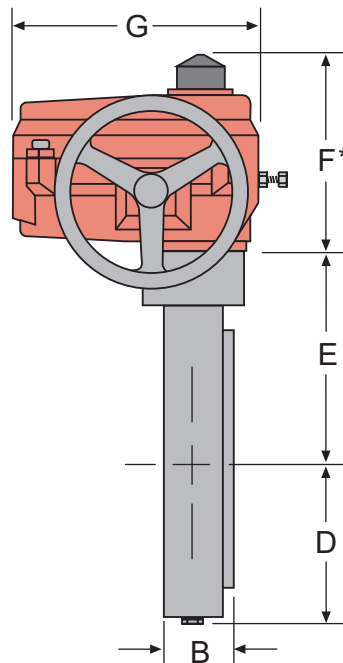
| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A     | B    | C     | D     | E     | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|-------|------|-------|-------|-------|------------------|-------|---------|
|              |              |                       |                       |       |      |       |       |       | BC               | Holes | Threads |
| 2-1/2"       | MKL2-X025    | 160                   | 78                    | 4.75  | 1.88 | 2.28  | 3.00  | 6.38  | 5.50             | 4     | 5/8-11  |
| 3"           | MKL2-X030    | 185                   | 123                   | 5.25  | 1.88 | 2.86  | 3.25  | 6.63  | 6.00             | 4     | 5/8-11  |
| 4"           | MKL2-X040    | 375                   | 250                   | 6.72  | 2.03 | 3.72  | 4.09  | 7.5   | 7.50             | 8     | 5/8-11  |
| 5"           | MKL2-X050    | 790                   | 360                   | 7.62  | 2.23 | 4.83  | 5.07  | 7.5   | 8.50             | 8     | 3/4-10  |
| 6"           | MKL2-X060    | 1350                  | 510                   | 8.62  | 2.23 | 5.88  | 5.57  | 8     | 9.50             | 8     | 3/4-10  |
| 8"           | MKL2-X080    | 2800                  | 1060                  | 10.81 | 2.40 | 7.80  | 6.94  | 9.5   | 11.75            | 8     | 3/4-10  |
| 10"          | MKL2-X100    | 4300                  | 1630                  | 13.06 | 2.75 | 9.78  | 8.56  | 10.75 | 14.25            | 12    | 7/8-9   |
| 12"          | MKL2-X120    | 6650                  | 2530                  | 15.42 | 3.08 | 11.74 | 10.18 | 12.25 | 17.00            | 12    | 7/8-9   |

↳ N=Normally Open / C=Normally Closed

FRONT VIEW



SIDE VIEW



| Actuator<br>Model | F*   | G     | H    |
|-------------------|------|-------|------|
| 70-0051           | 6.70 | 7.50  | 5.80 |
| 70-0081/0121/0201 | 8.10 | 10.10 | 7.80 |
| 70-0301/0501/0651 | 8.80 | 12.10 | 9.50 |

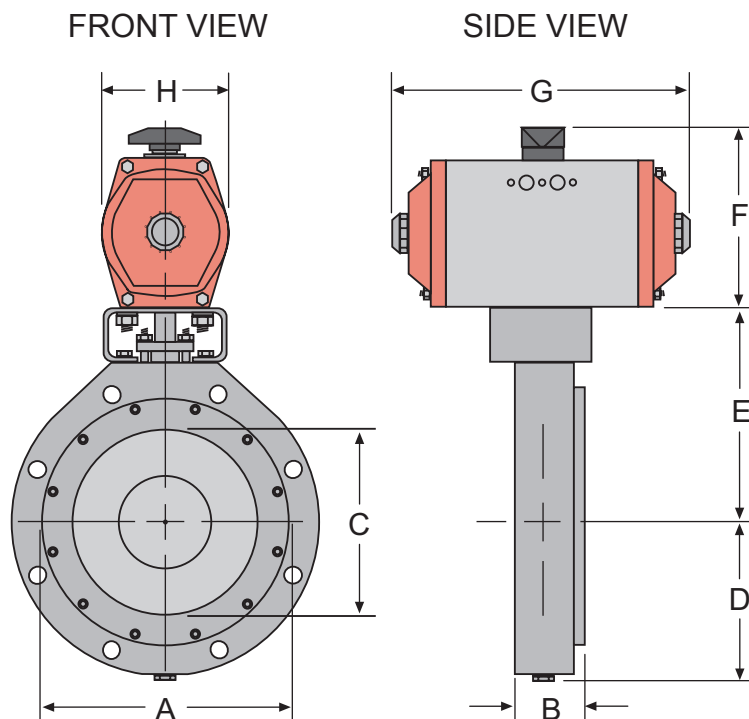
\* Allow additional 5" for Actuator  
Cover removal.

**CF Series**  
**Automated Butterfly Valves**  
**Dimensions**

**2-WAY SERIES 41 HIGH PERFORMANCE VALVE WITH SERIES 92 PNEUMATIC ACTUATOR**

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A     | B    | C     | D     | E     | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|-------|------|-------|-------|-------|------------------|-------|---------|
|              |              |                       |                       |       |      |       |       |       | BC               | Holes | Threads |
| 2-1/2"       | MKL2-X025    | 160                   | 78                    | 4.75  | 1.88 | 2.28  | 3.00  | 6.38  | 5.50             | 4     | 5/8-11  |
| 3"           | MKL2-X030    | 185                   | 123                   | 5.25  | 1.88 | 2.86  | 3.25  | 6.63  | 6.00             | 4     | 5/8-11  |
| 4"           | MKL2-X040    | 375                   | 250                   | 6.72  | 2.03 | 3.72  | 4.09  | 7.5   | 7.50             | 8     | 5/8-11  |
| 5"           | MKL2-X050    | 790                   | 360                   | 7.62  | 2.23 | 4.83  | 5.07  | 7.5   | 8.50             | 8     | 3/4-10  |
| 6"           | MKL2-X060    | 1350                  | 510                   | 8.62  | 2.23 | 5.88  | 5.57  | 8     | 9.50             | 8     | 3/4-10  |
| 8"           | MKL2-X080    | 2800                  | 1060                  | 10.81 | 2.40 | 7.80  | 6.94  | 9.5   | 11.75            | 8     | 3/4-10  |
| 10"          | MKL2-X100    | 4300                  | 1630                  | 13.06 | 2.75 | 9.78  | 8.56  | 10.75 | 14.25            | 12    | 7/8-9   |
| 12"          | MKL2-X120    | 6650                  | 2530                  | 15.42 | 3.08 | 11.74 | 10.18 | 12.25 | 17.00            | 12    | 7/8-9   |

→ N=Normally Open / C=Normally Closed



| Actuator<br>Model | F     | G     | H     |
|-------------------|-------|-------|-------|
| 92/93-63          | 4.29  | 6.00  | 2.76  |
| 92/93-83          | 5.20  | 7.89  | 3.58  |
| 92/93-92          | 5.54  | 8.91  | 3.94  |
| 92/93-118         | 6.93  | 12.10 | 4.72  |
| 92/93-127         | 7.73  | 12.34 | 5.39  |
| 92/93-160         | 9.37  | 15.45 | 6.77  |
| 92/93-210         | 11.61 | 18.92 | 8.83  |
| 90/91-255         | 13.46 | 26.70 | 10.75 |

## CF Series Automated Butterfly Valves

### Dimensions

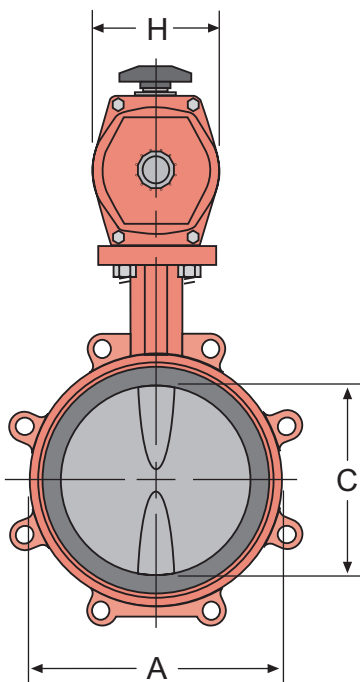
#### 2-WAY SERIES 31 VALVE WITH SERIES 92/93 PNEUMATIC ACTUATOR

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A     | B    | C     | D     | E     | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|-------|------|-------|-------|-------|------------------|-------|---------|
|              |              |                       |                       |       |      |       |       |       | BC               | Holes | Threads |
| 2"           | __L2-X020    | 144                   | 61                    | 3.69  | 1.62 | 2.00  | 2.30  | 5.50  | 4.75             | 4     | 5/8-11  |
| 2-1/2"       | __L2-X025    | 282                   | 107                   | 4.19  | 1.75 | 2.50  | 2.57  | 6.00  | 5.50             | 4     | 5/8-11  |
| 3"           | __L2-X030    | 461                   | 154                   | 4.88  | 1.75 | 3.00  | 2.81  | 6.25  | 6.00             | 4     | 5/8-11  |
| 4"           | __L2-X040    | 841                   | 274                   | 6.06  | 2.00 | 4.00  | 4.09  | 7.00  | 7.50             | 8     | 5/8-11  |
| 5"           | __L2-X050    | 1376                  | 428                   | 7.06  | 2.12 | 5.00  | 4.61  | 7.50  | 8.50             | 8     | 3/4-10  |
| 6"           | __L2-X060    | 1850                  | 567                   | 8.12  | 2.12 | 5.75  | 5.06  | 8.00  | 9.50             | 8     | 3/4-10  |
| 8"           | __L2-X080    | 3316                  | 1081                  | 10.50 | 2.50 | 7.75  | 6.05  | 9.50  | 11.75            | 8     | 3/4-10  |
| 10"          | __L2-X100    | 5430                  | 1710                  | 12.75 | 2.50 | 9.75  | 7.69  | 10.75 | 14.25            | 12    | 7/8-9   |
| 12"          | __L2-X120    | 8077                  | 2563                  | 14.88 | 3.00 | 11.75 | 9.02  | 12.25 | 17.00            | 12    | 7/8-9   |
| 14"          | __L2-X140    | 10538                 | 3384                  | 17.05 | 3.00 | 13.25 | 9.93  | 13.62 | 18.75            | 12    | 1-8     |
| 16"          | __L2-X160    | 13966                 | 4483                  | 19.21 | 4.00 | 15.25 | 11.30 | 14.75 | 21.25            | 16    | 1-8     |
| 18"          | __L2-X180    | 17214                 | 5736                  | 21.12 | 4.25 | 17.25 | 12.16 | 16.00 | 22.75            | 16    | 1 1/8-7 |
| 20"          | __L2-X200    | 22339                 | 7144                  | 23.25 | 5.00 | 19.25 | 14.00 | 17.25 | 25.00            | 20    | 1 1/8-7 |

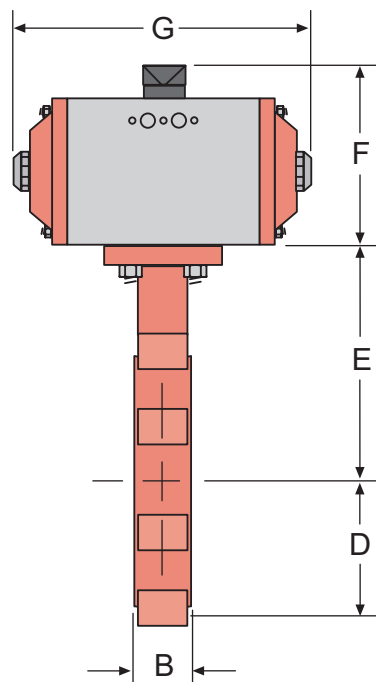
→ N=Normally Open / C=Normally Closed

→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

FRONT



SIDE VIEW



| Actuator<br>Model | F     | G     | H     |
|-------------------|-------|-------|-------|
| 92/93-63          | 4.29  | 6.00  | 2.76  |
| 92/93-83          | 5.20  | 7.89  | 3.58  |
| 92/93-92          | 5.54  | 8.91  | 3.94  |
| 92/93-118         | 6.93  | 12.10 | 4.72  |
| 92/93-127         | 7.73  | 12.34 | 5.39  |
| 92/93-160         | 9.37  | 15.45 | 6.77  |
| 92/93-210         | 11.61 | 18.92 | 8.83  |
| 90/91-255         | 13.46 | 26.70 | 10.75 |

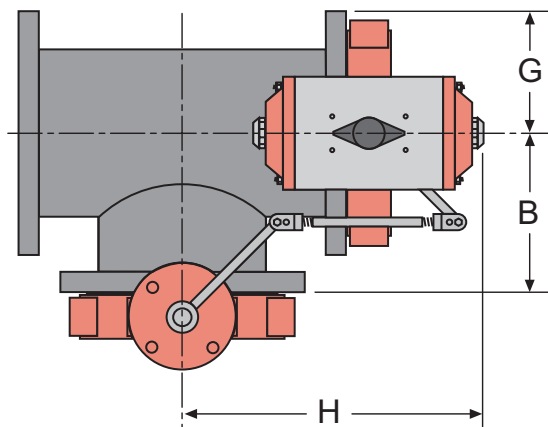
**CF Series**  
**Automated Butterfly Valves**  
**Dimensions**

**3-WAY SERIES 31 VALVE WITH SERIES 92/93 PNEUMATIC ACTUATOR**

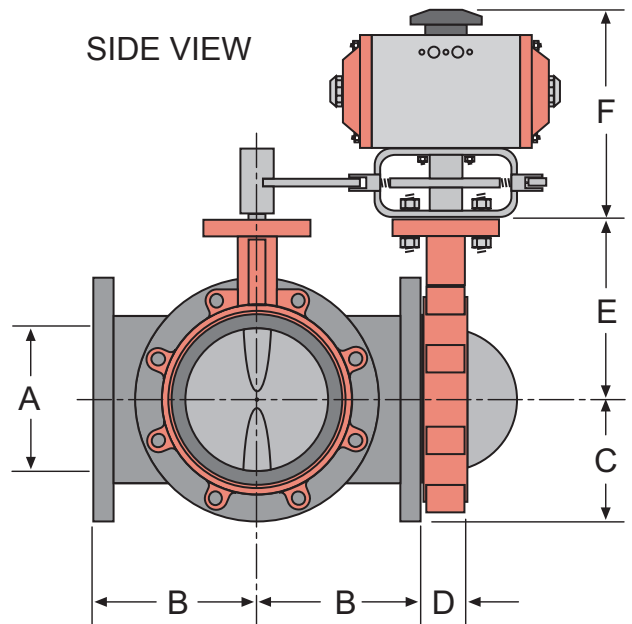
| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A     | B     | C     | D    | E     | Actuator<br>Model | F     | G     | H     |
|--------------|--------------|-----------------------|-----------------------|-------|-------|-------|------|-------|-------------------|-------|-------|-------|
| 2"           | _ _ L3-X020  | 144                   | 61                    | 2.00  | 4.50  | 3.00  | 1.62 | 5.50  | 92/93-063         | 7.30  | 3.00  | 8.30  |
| 2-1/2"       | _ _ L3-X025  | 282                   | 107                   | 2.50  | 5.00  | 3.50  | 1.80 | 6.00  | 92/93-083         | 8.20  | 3.50  | 9.80  |
| 3"           | _ _ L3-X030  | 461                   | 154                   | 3.00  | 5.50  | 3.80  | 1.80 | 6.20  | 92/93-083         | 8.20  | 3.80  | 10.30 |
| 4"           | _ _ L3-X040  | 841                   | 274                   | 4.00  | 6.50  | 4.50  | 2.00 | 7.00  | 92/93-083         | 8.20  | 4.50  | 11.40 |
| 5"           | _ _ L3-X050  | 1376                  | 428                   | 5.00  | 7.50  | 5.00  | 2.12 | 7.50  | 90/93-118         | 9.90  | 5.00  | 14.60 |
| 6"           | _ _ L3-X060  | 1850                  | 567                   | 5.80  | 8.00  | 5.50  | 2.12 | 8.00  | 92/93-118         | 9.90  | 5.50  | 15.20 |
| 8"           | _ _ L3-X080  | 3316                  | 1081                  | 7.80  | 9.00  | 6.80  | 2.50 | 9.50  | 92/93-127         | 11.70 | 6.80  | 18.40 |
| 10"          | _ _ L3-X100  | 5430                  | 1710                  | 9.80  | 11.00 | 8.00  | 2.50 | 10.80 | 92/93-160         | 13.40 | 8.00  | 19.90 |
| 12"          | _ _ L3-X120  | 8077                  | 2563                  | 11.80 | 12.00 | 9.50  | 3.00 | 12.20 | 92/93-210         | 15.60 | 9.50  | 23.00 |
| 14"          | _ _ L3-X140  | 10538                 | 3384                  | 13.20 | 14.00 | 10.50 | 3.00 | 13.60 | 92/93-210         | 17.60 | 10.50 | 25.00 |
| 16"          | _ _ L3-X160  | 13966                 | 4483                  | 15.20 | 15.00 | 11.80 | 4.00 | 14.80 | 90/91-255         | 19.50 | 11.80 | 30.40 |
| 18"          | _ _ L3-X180  | 17214                 | 5736                  | 17.20 | 16.50 | 12.50 | 4.20 | 16.00 | 90/91-255         | 19.50 | 12.50 | 32.00 |
| 20"          | _ _ L3-X200  | 22339                 | 7144                  | 19.20 | 18.00 | 13.75 | 5.00 | 17.20 | 90/91-255         | 19.50 | 13.80 | 33.80 |

→ Configuration Number  
→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

TOP VIEW



SIDE VIEW



## CF Series Automated Butterfly Valves

### Dimensions

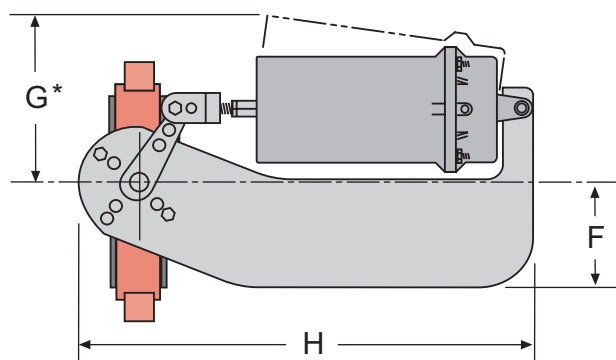
#### 2-WAY SERIES 31 VALVE WITH LOW PRESSURE PNEUMATIC ACTUATOR

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D    | E     | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|------|------|------|------|-------|------------------|-------|---------|
|              |              |                       |                       |      |      |      |      |       | BC               | Holes | Threads |
| 2"           | __L2-X020    | 144                   | 61                    | 2.00 | 2.30 | 1.62 | 5.50 | 8.62  | 4.75             | 4     | 5/8-11  |
| 2-1/2"       | __L2-X025    | 282                   | 107                   | 2.50 | 2.57 | 1.75 | 6.00 | 9.12  | 5.50             | 4     | 5/8-11  |
| 3"           | __L2-X030    | 461                   | 154                   | 3.00 | 2.81 | 1.75 | 6.25 | 9.38  | 6.00             | 4     | 5/8-11  |
| 4"           | __L2-X040    | 841                   | 274                   | 4.00 | 4.09 | 2.00 | 7.00 | 13.75 | 7.50             | 8     | 5/8-11  |
| 5"           | __L2-X050    | 1376                  | 428                   | 5.00 | 4.61 | 2.12 | 7.50 | 14.38 | 8.50             | 8     | 3/4-10  |
| 6"           | __L2-X060    | 1850                  | 567                   | 5.75 | 5.06 | 2.12 | 8.00 | 14.88 | 9.50             | 8     | 3/4-10  |
| 8"           | __L2-X080    | 3316                  | 1081                  | 7.75 | 6.05 | 2.50 | 9.50 | 16.38 | 11.75            | 8     | 3/4-10  |

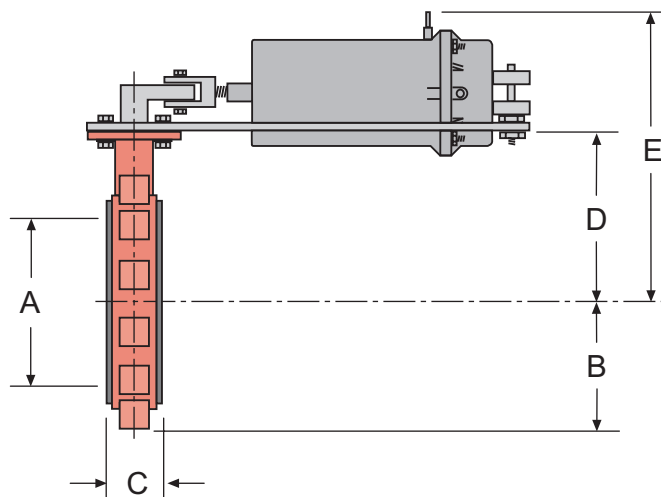
→ N=Normally Open / C=Normally Closed

→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

TOP VIEW



SIDE VIEW



| Actuator<br>Model | F    | G*   | H     |
|-------------------|------|------|-------|
| NR-D-3153         | 3.38 | 5.66 | 15.97 |
| NR-D-3244         | 4.88 | 7.31 | 19.94 |
| NR-D-3246         | 5.45 | 8.94 | 25.81 |

\* G = Max swing of actuator

**CF Series**  
**Automated Butterfly Valves**  
**Dimensions**

**2-WAY SERIES 31 VALVE WITH DUAL LOW PRESSURE PNEUMATIC ACTUATORS**

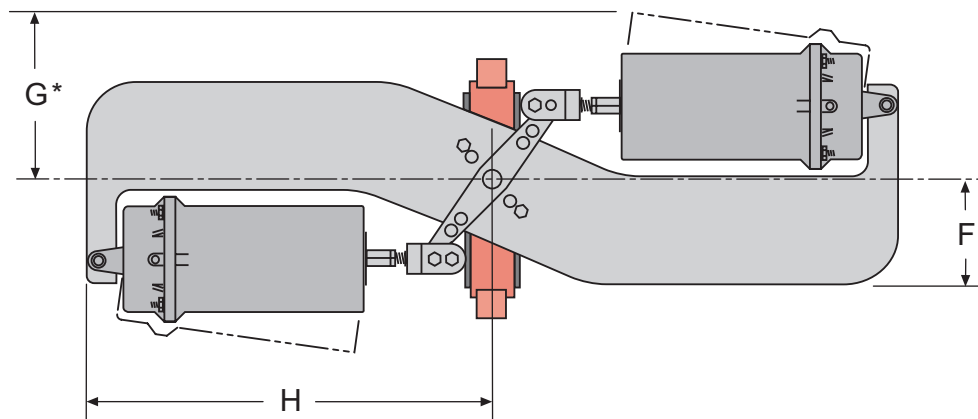
| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D     | E     | Lug Bolting Data |       |         |
|--------------|--------------|-----------------------|-----------------------|------|------|------|-------|-------|------------------|-------|---------|
|              |              |                       |                       |      |      |      |       |       | BC               | Holes | Threads |
| 8"           | L2-X080      | 3316                  | 1081                  | 7.75 | 6.05 | 2.50 | 9.50  | 16.38 | 11.75            | 8     | 3/4-10  |
| 10"          | L2-X100      | 5430                  | 1710                  | 9.75 | 7.69 | 2.50 | 10.75 | 17.62 | 14.25            | 12    | 7/8-9   |

→ N=Normally Open / C=Normally Closed  
→ NY=Nylon Coated Disc / AB=Aluminum Bronze Disc

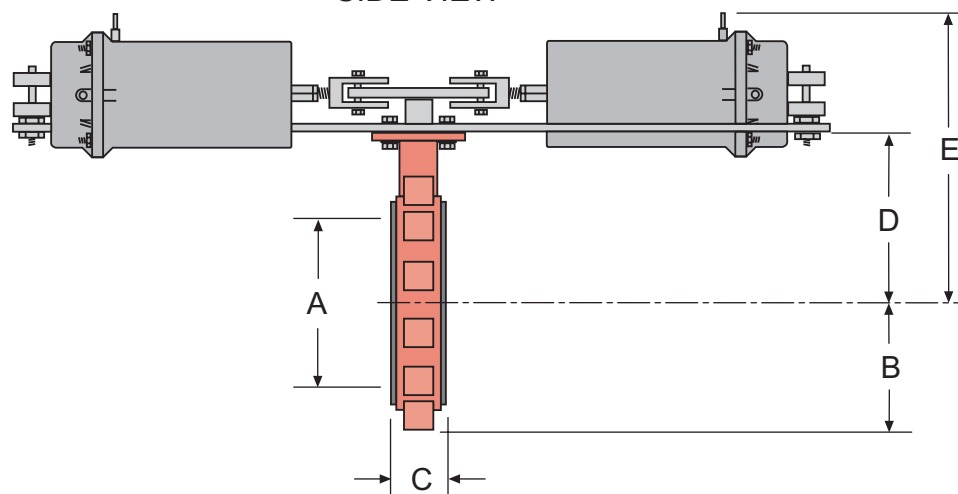
| Actuator<br>Model | F    | G*   | H     |
|-------------------|------|------|-------|
| NR-D-3246-D       | 6.00 | 8.94 | 22.75 |

\* G = Max swing of actuator

TOP VIEW



SIDE VIEW



## CF Series Automated Butterfly Valves

### Dimensions

#### 3-WAY SERIES 31 VALVE WITH LOW PRESSURE PNEUMATIC ACTUATOR

| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D    | E    |
|--------------|--------------|-----------------------|-----------------------|------|------|------|------|------|
| 2"           | _L3-x020     | 144                   | 61                    | 2.00 | 4.50 | 3.00 | 1.62 | 5.50 |
| 2-1/2"       | _L3-x025     | 282                   | 107                   | 2.50 | 5.00 | 3.50 | 1.80 | 6.00 |
| 3"           | _L3-x030     | 461                   | 154                   | 3.00 | 5.50 | 3.80 | 1.80 | 6.25 |
| 4"           | _L3-x040     | 841                   | 274                   | 4.00 | 6.50 | 4.50 | 2.00 | 7.00 |
| 5"           | _L3-x050     | 1376                  | 428                   | 5.00 | 7.50 | 5.00 | 2.12 | 7.50 |
| 6"           | _L3-x060     | 1850                  | 567                   | 5.80 | 8.00 | 5.50 | 2.12 | 8.00 |

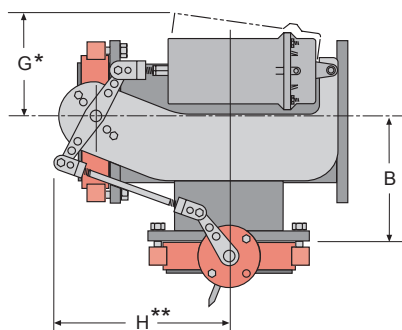
NY=Nylon Coated Disc AB=Aluminum Bronze Disc

| Actuator<br>Model | F     | G*   | H**   |
|-------------------|-------|------|-------|
| NR-D-3153         | 8.62  | 5.66 | 9.83  |
| NR-D-3153         | 9.12  | 5.66 | 10.40 |
| NR-D-3153         | 9.38  | 5.66 | 10.90 |
| NR-D-3244         | 13.75 | 7.31 | 12.02 |
| NR-D-3246         | 14.38 | 8.94 | 13.76 |
| NR-D-3246         | 14.88 | 8.94 | 14.26 |

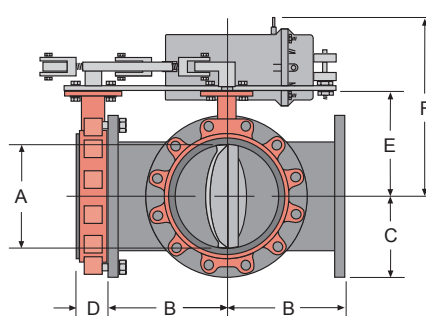
\* G = Max swing of actuator

\*\*H = Max swing of arm

TOP VIEW



SIDE VIEW



#### 3-WAY SERIES 31 VALVE WITH DUAL LOW PRESSURE PNEUMATIC ACTUATORS

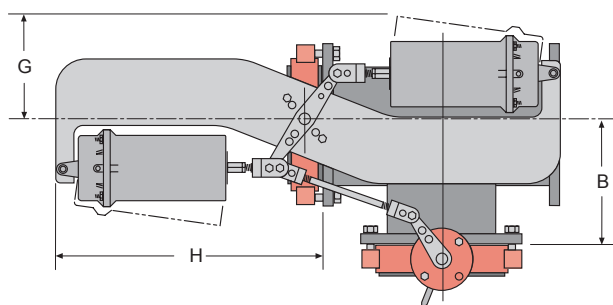
| Valve<br>NPT | Model<br>No. | C <sub>v</sub><br>90° | C <sub>v</sub><br>60° | A    | B    | C    | D    | E    |
|--------------|--------------|-----------------------|-----------------------|------|------|------|------|------|
| 6"           | _L3-x060     | 1850                  | 567                   | 5.80 | 8.00 | 5.50 | 2.12 | 8.00 |
| 8"           | _L3-x080     | 3316                  | 1082                  | 7.80 | 9.00 | 6.75 | 2.50 | 9.50 |

NY=Nylon Coated Disc AB=Aluminum Bronze Disc

| Actuator<br>Model | F     | G*   | H     |
|-------------------|-------|------|-------|
| NR-D-3246-D       | 14.88 | 8.94 | 23.81 |
| NR-D-3246-D       | 16.38 | 8.94 | 24.00 |

\* G = Max swing of actuator

TOP VIEW



SIDE VIEW

